Human-Computer Interaction

CSE 495 | Session 2020-21 | Final exam

Model study answers. Every listed option is addressed. Question titles are paraphrased; exact wording and original diagrams are in the source pages appended to this pack.

The uploaded HCI header says 4th Year 2nd Semester (session 2020-21). It is included here as requested, without relabeling the original paper.

Quick revision

Double Diamond: Discover -> Define -> Develop -> Deliver.

UCD loop: understand context -> specify requirements -> design -> evaluate -> iterate.

Experiment: independent variables are manipulated; dependent variables are measured.

Keyboard experiment: 2 sizes x 3 layouts x 3 device configurations = 18 combinations.

Usability measures: task success, completion time, errors and satisfaction.

1(a) | Why usability matters

Usability lets specified users achieve goals effectively, efficiently and with satisfaction in a specified context. It reduces errors, training effort and frustration, improving task completion and accessibility. Example: a clear course-registration flow helps students submit the right courses without repeated help.

Revision answer: Usability improves successful, efficient and satisfying task completion.

1(b) | Universal Design principles

The seven commonly taught Universal Design principles are equitable use; flexibility in use; simple and intuitive use; perceptible information; tolerance for error; low physical effort; and size and space for approach and use. Example: a campus kiosk offers readable text, audio alternatives, reachable controls and confirmation before destructive actions. These are distinct from Nielsen's usability heuristics.

Revision answer: Design for a broad range of abilities and contexts using the seven principles.

1(c) | What is WIMP?

WIMP means Windows, Icons, Menus and Pointer. It describes a familiar graphical-interface style where users manage windows, recognize icons, choose menu commands and point to controls. Example: a desktop file manager.

Revision answer: Windows, Icons, Menus, Pointer.

1(d) | What is the three-click rule?

It is a design heuristic claiming users should reach desired information within three clicks. Treat it as a historical guideline, not a validated universal limit: clear labels, meaningful progress and task success matter more than a fixed click count. A longer understandable path can be easier than three confusing clicks.

Revision answer: Keep navigation clear; three clicks is not a universal usability law.

1(e) | Claims and assumptions in conceptual design

An assumption is a belief taken as true without yet having evidence, such as assuming all students own smartphones. A claim is an assertion about an expected design benefit, such as saying push reminders will reduce missed deadlines. Record both and test them through research or evaluation before relying on them.

Revision answer: Assumptions are untested beliefs; claims are assertions to investigate.

1(f) | Mind maps

A mind map is a visual organization of ideas around a central topic. Branches show related concepts and sub-branches show detail. For a study app, branches might include courses, notes, revision and notifications. It supports brainstorming and seeing relationships rather than prescribing a rigid workflow.

Revision answer: A central idea with branching related concepts.

1(g) | Problems with interface metaphors

A metaphor can imply incorrect behavior, transfer poorly across cultures, become confusing when pushed beyond its real-world meaning, or constrain better designs. A trash-can metaphor may confuse users about whether an item is recoverable. Use metaphors only where they clarify the system and support them with clear labels and feedback.

Revision answer: Metaphors help familiarity but can mislead or limit understanding.

1(h) | Wizard-of-Oz prototyping

Users interact with what appears to be an automated system while a human secretly performs some system responses. A researcher might simulate a café chatbot before building its AI. It tests interaction ideas cheaply, but timing and behavior must be consistent, and research ethics/consent and debriefing should be handled appropriately.

Revision answer: A human simulates functionality behind an apparently working prototype.

2(a) | Four stages of the Double Diamond

Discover: explore users and the problem broadly through interviews, observation and evidence. Define: synthesize findings into a clear problem statement. Develop: generate and prototype multiple solutions. Deliver: test, refine and implement a suitable solution. Each diamond expands possibilities and then narrows them. Example: investigate café queues, define pickup confusion, prototype alternatives, then test and launch an improved order flow.

Revision answer: Discover, Define, Develop, Deliver; diverge then converge twice.

2(b) | Nine user-centered chapters

The wording appears to refer to a course-specific nine-step framework, but no lecture framework is attached. One defensible nine-stage study outline is: identify users and stakeholders; research context; understand goals and tasks; specify user requirements; generate concepts; structure information and flows; prototype; evaluate with users; iterate and monitor after release. Explain how each keeps user evidence central. Do not present this particular nine-step enumeration as a uniquely standardized UCD model; compare its labels with your lecture notes.

Revision answer: A nine-stage example is provided; the exact course-specific list needs the lecture source.

2(c) | Slow, hard-to-navigate university portal

Slow responses harm efficiency and satisfaction; poor navigation harms learnability, findability and task success, and can increase errors. Measure completion time, success rate, navigation mistakes and user feedback. Improve information architecture, use consistent plain-language labels and visible navigation, show loading/progress feedback, optimize server and page performance, and test common tasks such as registration with students.

Revision answer: Improve performance and findability, then verify using usability measures.

2(d) | Desktop, folders and trash can

This is an interface metaphor: familiar real-world concepts help users form a mental model of an unfamiliar digital system. A folder suggests grouping documents; a trash can suggests discarding items. This reduces initial learning effort, but the design should explain differences such as restoration and permanent deletion.

Revision answer: An interface metaphor transfers familiar concepts into a digital interface.

2(e) | Organizing brainstorming sticky notes

Use affinity diagramming. Put each observation or idea on a separate note, silently group related notes, discuss and adjust the groups, label each cluster with a meaningful theme, and inspect relationships or higher-level themes. Preserve unusual notes rather than forcing everything into a category. Example clusters: navigation, payment, accessibility and feedback.

Revision answer: Affinity diagramming groups related ideas into labeled themes.

3(a) | Low- and high-fidelity café app prototypes

A prototype is a representation used to explore and test a design before committing to a full implementation. Low fidelity: paper sketches of menu, cart, checkout and pickup; ask users to order while a facilitator switches screens. It is cheap, quick and good for comparing flows. High fidelity: a realistic clickable interface with representative menu data, order confirmation and simulated payment states; test labels, interaction details and error recovery. It costs more and can make unfinished features appear real. Compare what the evaluation needs to learn, then iterate; do not collect real card details in a simulated payment test.

Revision answer: Use low fidelity for early structure and high fidelity for realistic interaction testing.

3(b) | Primary interaction types and examples

A commonly taught interaction-design set is instructing, conversing, manipulating and exploring. Instructing: choose Print or issue a command. Conversing: ask a voice assistant a question. Manipulating: drag a file or pinch a photo. Exploring: navigate a map or virtual environment. Some newer course frameworks also discuss responding to system prompts as an additional type; use your lecture's taxonomy if it differs.

Revision answer: Instructing, conversing, manipulating and exploring, with course-dependent extensions.

3(c) | UCD process for a mobile banking app

Understand the context: interview customers, including those with accessibility needs, about devices, connectivity and banking tasks. Specify requirements: safe transfers, clear balances, recovery and measurable usability goals. Produce design solutions: prototype account views, recipient selection, amount entry and confirmation. Evaluate with representative users using realistic tasks and safe test data. Iterate based on failures and evidence, then monitor after release. Security and usability requirements must be considered together.

Revision answer: Understand users -> specify requirements -> design -> evaluate -> iterate.

4(a) | Ten-second rule in web design

The common guidance is that users should quickly understand a page's purpose and value; after roughly ten seconds without engagement or useful feedback, attention may drift. It is a heuristic, not a hard universal time limit. Give a clear heading, relevant content and responsive progress feedback; do not imply that every task must finish in ten seconds.

Revision answer: Communicate purpose and progress quickly; ten seconds is a guideline.

4(b) | Issues with long URLs

Long, opaque URLs are hard to remember, read, type and share, may wrap or be truncated, and can hide meaningful structure. Query strings can also accidentally expose sensitive information. Prefer stable, descriptive routes and avoid putting secrets in URLs. A long URL does not automatically make the underlying page unusable.

Revision answer: Long URLs reduce readability and shareability and can expose query data.

4(c) | Think-aloud limitations

Speaking while working can slow participants, change their strategy and impose cognitive load. Some participants find it unnatural or cannot verbalize automatic behavior; retrospective explanations can be incomplete. Use neutral prompts, avoid coaching, and distinguish task-time measurements gathered under think-aloud from normal performance.

Revision answer: Verbalization can alter the behavior being measured.

4(d) | Gathering subjective data

Use interviews, questionnaires, satisfaction scales, diaries and focus groups to collect perceptions and experiences. For example, a post-task ease rating measures perceived difficulty, while an interview explores why. Combine subjective reports with observed behavior; stated preference and actual success can differ.

Revision answer: Interviews, questionnaires, scales, diaries and focus groups capture user perceptions.

4(e) | Why informed consent matters

Participants should understand the study's purpose, procedures, expected risks, recording and data use, confidentiality arrangements and right to withdraw. Consent respects autonomy and supports ethical participation. Avoid coercion and collect only necessary data. Consent is a process, not merely a signature.

Revision answer: People must be able to make an informed, voluntary decision about participation.

4(f) | Heuristic evaluation

Usability specialists inspect an interface against established principles such as visibility of status, consistency and error prevention. Evaluators first inspect independently, then consolidate findings and severity judgments. It is inexpensive early feedback but does not replace testing with representative users.

Revision answer: Expert inspection against usability principles.

4(g) | Predictive evaluation methods

Predict performance without running a full user experiment using models such as GOMS or the Keystroke-Level Model. Fitts's law can estimate pointing time from target distance and size, and Hick-Hyman-style models can describe choice-time effects under suitable assumptions. Predictions depend on model assumptions and should be checked against real tasks.

Revision answer: Model-based evaluation includes GOMS, KLM and suitable pointing/choice models.

4(h) | People-centered approach

Design around people's goals, capabilities, needs and real context, involving affected people throughout research and iteration. Consider diverse users and stakeholders rather than only technical convenience. Example: test a service with first-time users and people with disabilities, not just its developers.

Revision answer: Let evidence about people guide design and evaluation.

5(a) | Smart speaker fails in background noise

This illustrates context-sensitive speech recognition and input ambiguity. Test in realistic acoustic conditions; improve noise handling and microphone placement; show or speak back recognized commands; ask for confirmation on consequential actions; allow correction and provide touch/text alternatives. Avoid repeatedly blaming the user for recognition failures.

Revision answer: Support noisy contexts with feedback, recovery and alternative inputs.

5(b) | Mandatory account creation before checkout

A forced registration barrier adds unnecessary effort and limits user freedom, harming flexibility and efficiency. Offer guest checkout, request only essential information, explain why it is needed and allow optional account creation after purchase. Measure abandonment and task completion before and after changing the flow. There is no single universally named heuristic uniquely implied by this scenario.

Revision answer: Guest checkout reduces unnecessary effort and preserves user choice.

5(c) | HCI principles for mobile and web

Use consistent navigation, readable hierarchy, clear status, error prevention/recovery and accessible controls. Small screens require prioritization, responsive layouts and progressive disclosure. Touch needs sufficiently large, separated targets and alternatives to hidden gestures; avoid hover-only access. Web layouts should support keyboard navigation, zoom, varied screen sizes and visible focus. Optimize loading and preserve task state across interruptions. Test on representative devices and assistive technologies.

Revision answer: Design for screen size, input method, accessibility, responsive layout and feedback.

5(d) | Experiment: does a new UI reduce transaction time?

Define a hypothesis and primary outcome, such as successful transaction completion time. Independent variable: old versus new interface. Recruit representative users and assign conditions randomly in a between-subjects design, or counterbalance order in a within-subjects design. Keep tasks, device, network and instructions comparable. Pilot the procedure; record time, errors and success rather than ignoring failed tasks. Compare conditions with an appropriate analysis, report uncertainty/effect size and consider practical significance. Control practice and fatigue effects.

Revision answer: Manipulate interface version, control confounds and compare timed task performance.

5(e) | Ethnographic study of older adults and smart homes

Observe participants in their normal homes over time, with informed consent and privacy protections. Use contextual observation, open interviews, field notes, participant diaries and device-interaction records where consented. Study routines, physical constraints, family support, workarounds, trust and breakdowns in context. A lab may miss shared-device practices or connectivity problems, while surveys may miss behavior participants take for granted. Analyze recurring patterns without treating all older adults as identical.

Revision answer: Study actual routines and workarounds in the home through observation and contextual inquiry.

6(a) | Five steps of usability testing for e-commerce

1. Plan: define target users, research questions, metrics and representative tasks such as finding and buying an item. 2. Recruit: select representative participants and arrange accessible, ethical participation. 3. Prepare: create a realistic test environment, safe test accounts, tasks and a pilot. 4. Conduct: give neutral tasks, observe without coaching, and record success, time, errors and comments. 5. Analyze and improve: group problems, prioritize by severity/frequency, revise and retest. This is a practical five-step organization; textbooks may divide these activities differently.

Revision answer: Plan, recruit, prepare, conduct, analyze and improve.

6(b) | Hypotheses and keyboard experiment variables

A hypothesis is a testable statement about a population or effect. H0 usually expresses no difference/effect; H1 expresses a difference, or a specified direction if defined in advance. Here H0 states no performance or preference differences across the keyboard conditions. Independent variables: keyboard size (small, large: 2 levels); layout (alphabetic, QWERTY, random: 3); device/input configuration (touch large screen, mouse monitor, stylus PDA: 3). A full crossing gives 2*3*3=18 conditions. Dependent variables: completion/typing time, error rate and preference on a 5-point Likert scale. Control variables might include text length/difficulty, instructions and environment. Counterbalance order to limit learning/fatigue. Because device and input method are bundled, their separate causal effects cannot be isolated from this design alone.

Revision answer: Three factors with 2,3,3 levels; measure time, errors and preference; control other influences.

6(c) | Focus groups and effective interviews

A moderator gathers a small, relevant group to discuss experiences using an open discussion guide. Obtain consent, encourage quieter participants, prevent domination, probe disagreements and record/analyze themes. Use findings to refine interpretations, then triangulate with observation or individual interviews; group agreement alone is not proof. Effective interviews use neutral open questions, ask for concrete recent examples, probe gently, listen actively, avoid leading or double-barreled questions and summarize interpretations for clarification.

Revision answer: Moderate balanced group discussion and use neutral, evidence-seeking interview questions.

Reference reading

Yvonne Rogers, Helen Sharp and Jenny Preece, Interaction Design: Beyond Human-Computer Interaction.

Alan Dix, Janet Finlay, Gregory D. Abowd and Russell Beale, Human-Computer Interaction.

Jakob Nielsen, 10 Usability Heuristics: https://www.nngroup.com/articles/ten-usability-heuristics/

These are supporting reading references, not claims that this paper or its answer key appears in those books. Course-specific terminology and source ambiguities are noted beside the affected answers.

Shahjalal University of Science and Technology
4th Year 2nd Semester Final Exam (Session 2020-21)
Course Title: Human Computer Interaction.
Course Code: CSE 495, Credit: 3.0
Marks: 100; Time: 3 hours

Note: Answers all questions, Use single script, please answer the questions in order. Possession of mobile phones is strictly prohibited

Part A

1 ***Answer any 5*** 5*2

a) Why is usability important?
b) Mention Universal Design Principal.
c) What is WIMP interface?
d) What is 3 click rules?
e) What is claim and assumption in conceptualizing the design?
f) What is mind maps?
g) What are the problems of interface metaphors?
h) What is 'Wizard-of-Oz' prototyping?

2 ***Answer any 4*** 4*5

a) Imagine you're on a journey to solve a big design problem. The "Double Diamond" is your map. Can you explain the four main territories you travel through on this map to get from the problem to the perfect solution?

b) In the story of creating a great product, the user is the hero. What are the nine key chapters in this "user-centered" story that make sure we always keep the focus on our hero from start to finish?

c) Students complain that university portal is slow and difficult to navigate. Which usability aspects are being affected. Propose a solution.

d) On our computers, we have a "desktop," "folders," and a "trash can" to help us understand how to use them without a manual. What is this clever trick of using familiar, real-world ideas called, and why is it so powerful for making new technology feel easy?

c) Picture this: you've just finished a brainstorming session and have a wall full of scattered sticky notes. What is the magical technique called that helps you sort all this chaos into neat, organized groups, and how do you perform this trick?

3 ***Answer any 2*** 2*10

a) Imagine you are a user experience (UX) designer creating a new mobile app for a local cafe that allows customers to order and pay ahead.
Given this scenario, **what is prototyping, and how would you use both low-fidelity and high-fidelity prototyping to test and refine the app's design?** What are the key differences between these two types of prototypes in this context?

b) What are the primary types of interactions in user experience (UX) design, and how would you apply each to a real-world example

c) What are the core steps of the **User-Centered Design (UCD)** process? Briefly explain how a product team might apply these steps to develop a mobile banking app.

Part B

4 ***Answer any 5*** 5*2

a) What is 10 second rule in Web design?
b) What is the issue of long URL?
c) What is the potential problem of *Think a loud* approach?
d) What are the methods of gathering subjective data?
e) Why is consent important for the participants before joining experiment?
f) Define heuristic evaluation.
g) What are the methods of predicative evaluation?

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h) What is a a people centered approach?

5 ***Answer any 4*** 4*5

a) A smart speaker often misinterprets voice commands due to background noise. What challenge does this illustrate? Suggest solution to improve user experience.

b) An ecommerce website asks users to create an account before checkout, leading many to abandon their carts. Which usability principle is ignored? Suggest alternative design to reduce the drop-offs.

c) Explain the key HCI design principles for mobile and web interfaces. Discuss the challenges like screen size, navigation, touch gesture and responsiveness.

d) A software company wants to know if a new user interface reduces the time taken to complete online transactions. Describe the key procedures in experimental design, showing how they would apply to this investigation.

e) Imagine a technology company wants to understand how elderly adults interact with smart home devices.
Given this scenario, **how would an ethnographic study be used to collect data on their user experience?** What specific methods would a researcher use to gather insights that couldn't be obtained through traditional surveys or lab tests?

6 ***Answer any 2*** 2*10

a) How would you describe the five basic steps of usability testing to a new product manager? For example, if you were testing a new e-commerce website, what would you do at each step to ensure it is user-friendly and effective?

b) What is Hypothesis? What is the purpose of Null and Alternate hypothesis?

Hypothesis: There is no difference in user performance (time and error rate) and preference (5 point likert scale) when typing on two sizes (i.e., small, large) of an alphabetic, qwerty and random on-screen keyboard using a touch-based large screen, a mouse- based monitor, or a stylus-based PDA.

What are the independent variables here? What are the levels? What are the dependent variables here? What is the purpose of control variable?

c) How can a researcher use a focus group discussion (FGD) to collect and validate data, and what are some effective interview techniques for data collection?

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